Kanageshwaran
Stat 50

Homework #5
Section : 2.2

(2) fluxes 4
base metals 2
energy input 3

ways = 4 · 2 · 3 = 24

∴ There are 24 different ways for each have exactly one runs.

(3) # different choices = $\frac{8!}{4!} = {}^8C_4 = 70$

∴ There are 70 different choices were there for the four locations to apply the new ointment.

(4) ⇒ order doesn't matter
⇒ 18 choose 9
⇒ then, # way = ${}^{18}C_9 = 48620$

∴ There are 48620 ways to do it.

(5) (a) ⇒ order does matter
then, # ways = ${}^8P_3 = 336$.
∴ There are 336 ways.

(b) ⇒ Order doesn't matter

then, # ways = $_8C_3 = 56$

∴ There are 56 ways to

(6) $n = 10$

⇒ Order does matter

⇒ 10 choose 3

⇒ then, # ways = $_{10}P_3 = 720$

∴ There are 720 ways.

(7) $n = 15$

10 → 2 choices

5 → 4 choices

Let's divide into 2 sums:

1) true or false : # way = $2^{10} = 1024$

2) multiple choice : # ways = $4^5 = 1024$

∴ total Number of ways = $1024 \times 1024 = 1048576$

ways = 1048576

(8)

(a) 3 → letters → 26 choices

3 → digits → 10 choices

Then, # ways = $26^5 \cdot 10^3 = 17576000$.

(b) 3 → letters without Q → 25 choices
3 → digits without 9 → 9 choices

Then, # ways = $25^3 \cdot 9^3 = 11390625$

(c) P(C) = $\frac{\text{\# ways plates can be made without "Q" and "9"}}{\text{\# ways total plate can be made.}}$

P(C) = $\frac{11390625}{17576000} = 0.648078345$

P(C) = 0.6481

(9)

(a) 8 character → 36 choices
then, # ways = $36^8 = 2.8211 \times 10^{12}$

(b) 8 character → 36 choices, but not includes 8 characters only with 26 choices of lower case.
Then, # ways = $36^8 - 26^8 = 2.6123 \times 10^{12}$

(c) $P(C) = \frac{\text{\# ways that password can be created with at least 1 control.}}{\text{\# ways of password can be made in total}}$

$$P(C) = \frac{36^8 - 26^8}{36^8} = \frac{2.6123 \times 10^{12}}{2.8211 \times 10^{12}}$$

$$\boxed{P(C) = 0.9260}$$

(11)

A			B		
color	number	Prob	color	numbers	
blue	8	8/14	blue	4	4/6
white	6	6/14	white	2	2/6

0.8025238

$P(A)$ = choosing blue in A and B

$P(B)$ = choosing white in A and B

$$P(A) = \frac{8}{14} \cdot \frac{4}{6} = \frac{32}{84}$$

$$P(B) = \frac{6}{14} \cdot \frac{2}{6} = \frac{12}{84}$$

$$P(A) + P(B) = \frac{32}{84} + \frac{12}{84} = \frac{44}{84}$$

$$P(A) + P(B) = 0.523809524$$

∴ The probability of getting matching socks is 0.52381

(12)

color	numbers
red	6
green	4
black	2

Chosing 2 red :-

$P(R) = \frac{6}{12} \cdot \frac{5}{11} = 0.227272727$

Chosing 2 green :-

$P(G) = \frac{4}{12} \cdot \frac{3}{11} = 0.090909091$

Chosing 2 black :-

$P(B) = \frac{2}{12} \cdot \frac{1}{11} = 0.015151515$

$P(\text{Match}) = P(R) + P(G) + P(B)$.

$= 0.227272727 + 0.090909091 + 0.015151515$

$= 0.333333333$

$\boxed{P(M) = 0.3333}$